

Paul He

☎ 416 434 2761

✉ paul.he@mail.utoronto.ca

🌐 [linkedin](#)

🔄 GitHub

🌐 Academic Website

Education

University of Toronto

September 2021 - May 2025

Honours Bachelor of Science in Computer Science

Toronto, Canada

- Enrolled in Arts & Science Co-Op program
- Dean's List
- **Relevant Coursework:** Introduction to Machine Learning

Swiss Federal Institute of Technology (ETH Zürich)

September 2023 - August 2024

Exchange Year

Zürich, Switzerland

- EESTEC Generative AI hackathon winner
- **Relevant Coursework:** Visual Computing, Natural Language Processing (Graduate), Probabilistic Artificial Intelligence (Graduate), Machine Perception (Graduate), Large Language Models (Graduate), Advanced Formal Language Theory (Graduate)

Research

Word Segmentation and Part-of-Speech Tagging with Transformer

October 2023 – January 2024

Supervisors: Tianyu Liu, Prof. Dr. Ryan Cotterell Paper

ETH Zürich

- Designed a **new decoder embedding mechanism** allowing the decoder to gain access to finer-grained features, enabling it to better capture subtle linguistic nuances and dependencies between characters and tags.
- Implemented a **transformer** model using **PyTorch**, beam-search for generation.
- Achieved leading-edge results, demonstrating the model's efficiency with a validation accuracy of **96.29%** for CWS, **93.37%** for POS tagging, and **91.99%** for joint CWS and POS tagging after just 10 epochs.

Projects

Mental Health Support Bot | Python, beautifulsoup4, FastAPI

October 2023

- Led the development of simple mental health chatbot web-app tailored for ETH Computer Science students, assisting team members in debugging.
- Performed **web scraping** to fine-tuned pre-trained language model.
- Implemented a real-time chat feature FastAPI to host our chat platform and model on a server, allowing multiple chat's with the bot at once, and also recalls previous conversations

Game Matchup Estimator | Python, Pytorch, Numpy, scikit-learn, Flask, HTML, CSS

May 2023 - September 2023

- Integrated the Clash Royale API to retrieve real-time game data, generating up to 130,000 labeled data points.
- Developed a neural network model with **PyTorch** to analyze large-scale data and accurately predict match outcomes with 73% accuracy.
- Utilized techniques such as **regularization, and hyperparameter tuning** to enhance the model's predictive capabilities and mitigate overfitting.
- Wrote a simple web interface using **Flask, HTML, and CSS** to visualize and interact with the system's predictions.

Clinic Management System | Java, JUnit, Git

June 2022 - August 2022

- Created a Clinic Management System adhering to **Clean Architecture** and **SOLID principles**, while incorporating **Design Patterns** for optimal code structure.
- Collaborated with a **team of 8**, leveraging **Git & Github** for efficient version control, pull requests, issue tracking, and code reviews.
- Ensured code reliability through comprehensive test coverage by developing robust test suites using the JUnit framework.

Technical Skills

Programming Languages: Python, Java, C, R, SQL

Frameworks/Libraries: PyTorch, scikit-learn, NumPy, FastAPI, Flask

Tools/Platforms: Git, Docker, Jupyter, GitHub

Databases: PostgreSQL, psycopg2